

Lalit Negi, M.Tech

Member Research Staff (MRS) at Bharat Electronics Limited

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BIOGRAPHY

My primary research interests lie at the intersection of Geographic Information Systems (GIS), simulation, Machine Learning (ML) and Augmented/Virtual Reality (AR/VR). I am particularly interested in geospatial modeling and simulation, 3D GIS, immersive geospatial visualization, spatial analysis, and the integration of GIS with AR/VR technologies for interactive and realistic representations of complex environments. I am also interested in exploring how simulation and immersive technologies can support applications such as spatial decision-making, urban and environmental modeling, digital twins, and interactive geospatial systems. Previously, I have also worked in Applied Cryptography and Information Security, which has given me a background in secure systems and data protection.

I am particularly interested in PhD research at the intersection of GIS, spatial simulation, GeoAI, and human-environment systems, with a focus on developing inclusive technologies for marginalised groups, including LGBTQ+ communities, and understanding how spatial, social, and environmental factors shape their experiences over time.

EXPERIENCE

Central Research Laboratory, Bharat Electronics Limited

Member Research Staff

India, Ghaziabad

2022 – Present

GIS Tech Lab

Jan 2024- Present

- Currently working on web based **CIDSS** Command Information and Decision Support System for the Indian Army using the **Luciad RIA sdk package**.
- Leading the team working on the development of **VR Based Digital Sandtable** prototype using the **ArcGIS sdk** in Unity
- Designed a virtual Tactical Theatre Command system using **ArcGIS**, visualizing real-world deployment of defence resources, operational assets, and live tracks to support situational awareness and command-level planning.
- Successfully participated in the demonstration of the **Tactical Theatre Command System** to the **Honourable Defence Minister of India**, showcasing real-time visualization, operational planning, and decision-support capabilities.

Decision Support System Lab

July 2022 - Dec 2023

- Contributed to the development of the **first indigenously developed universal train driving simulator** through simulation algorithm design and secure communication protocol development.
- Developed simulation modules such as **Simulation Engine**, **Training and Evaluation Assessment Module**, and **Malfunction Handler** in **C++** using **Qt** to model metro train logic, control, and operational behavior.
- Prepared technical documentation including **Software Requirements Specification (SRS)** and **Software Requirement Description (SRD)** to support project delivery and audits.
- Developed a **VR-based training simulator** for radar assembly and disassembly, enabling step-by-step procedural training in an immersive and safe virtual environment.
- Developed a **SR-based display system** to create a **Digital twin** of radar systems for predictive maintenance, condition monitoring, and fault analysis.

PERSONAL PROJECTS

• Rainbow Connect

- Theme: A web-based communication platform designed to support LGBTQ+ communities in offices and institutions. The platform enables users to connect through text-based communication and video calling, with support for both Hindi and English. Designed as an inclusive digital space to encourage communication, connection, and peer interaction within institutional environments.

• Indradhanush Spaces

- Theme: A web-based GIS application for discovering and registering LGBTQ+-inclusive spaces. Space owners can register their inclusive spaces on the platform, while users can explore nearby spaces through an interactive map. The application uses location-based visualization to display spaces around the user's current location and identify places within a selected geographic range.
- **Geographic Information System-Based Real-Time Fire Detection and Anti-Fire Simulation System**
 - Theme: A GIS-based system for real-time fire detection, spatial visualization, and fire-spread simulation. Integrates geospatial data, real-time information, and simulation models to support fire monitoring, prediction, and emergency response planning.
- **Interactive Radar Fault Management Training System with GIS-Based Visualization and Corrective Action Guidance**
 - Theme: An interactive training system that integrates GIS-based visualization with radar fault scenarios and corrective-action guidance. Designed to provide an immersive and interactive environment for fault identification, decision-making, and technical training.
- **Immersive Virtual Reality Visualization of Machine Learning-Based Weather Classification for Climate Scenario Analysis**
 - Theme: An immersive virtual reality environment for visualizing machine-learning-based weather classification results and exploring climate scenarios. Combines VR, geospatial visualization, and machine learning to provide an interactive approach to understanding complex weather and climate data.

EDUCATION

- **Netaji Subhas University of Technology** 2020 – 2022
Master of Technology (M.Tech.) in Information Technology
 - Thesis Title: Design and Analysis of Certificateless Schemes for Healthcare Environment
 - Aggregate: 81.0% (GPA 8.1/10.0)
- **Govind Ballabh Pant University of Agriculture and Technology** 2017 – 2020
Bachelor of Technology (B.Tech.) in Computer Engineering
 - Project Title: Steg! - An Android Application for Metamorphic Cryptography
 - Aggregate: 72.4% (GPA: 7.2/10.0)

PERSONAL SKILLS

Technical Skills	◦ Domain: Inclusive Technology, GIS, ARVR, Machine Learning ◦ Languages: C#, Python, C++, TypeScript ◦ Tools: Unity, React, Vite, Canvas, Latex, Github
Communication Skills	◦ Assisted in conducting laboratory sessions and tutorials, supporting students in understanding course concepts and practical implementations. ◦ Guided students with assignments, mini-projects, and problem-solving exercises. ◦ Helped in preparing teaching materials such as lecture slides, lab manuals, and tutorial sheets. ◦ Mentoring Interns in the Integrated Domain of Machine Learning and Virtual Reality.
Organizational Skills	◦ Served as an organizer for a five-day technical workshop sponsored by All India Council for Technical Education, coordinating planning, scheduling, and execution of technical sessions. ◦ Assisted in the preparation of workshop materials, registrations, and participant communications. ◦ Acted as an organizing member for the inauguration event of a DPSU Bhawan, supporting event planning and on-site coordination. ◦ Coordinated ceremonial arrangements, guest management, and technical arrange-

ments for the inauguration event.

Certifications

- **Geographical Information System** by Department of Space, India
- **India in a Changing Climate** by Indian Institute of Technology Hyderabad, India.
- **Artificial Intelligence for Social Good** by Microsoft.

PUBLICATIONS

- Presented **5+ papers** in International and national conferences
- Published **2 papers** in SCI indexed Journals.
- GIS Based Real time Fire Detection and Anti Fire Simulation System.(Accepted, 47th Asian Conference on Remote Sensing (ACRS) 2026)

REFERENCES

1. **Shivani Arya**

Senior Member Research Staff, CRL-BEL, Ministry of Defence, India

Email: shivaniarya@bel.co.in

Relationship: Reporting Manager

2. **Dr. Devender Kumar Yadav**

Associate Professor, NSUT, New Delhi, India

Email: dk_iitm@yahoo.co.in

Relationship: Master's Advisor